Strategies for Transforming Multiscale and Multimodal Image Data to a Common Format for Cloud Integration and Enhanced Visualization

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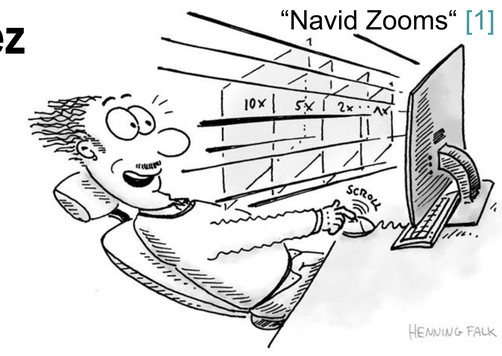


"Navid Zooms" [1]

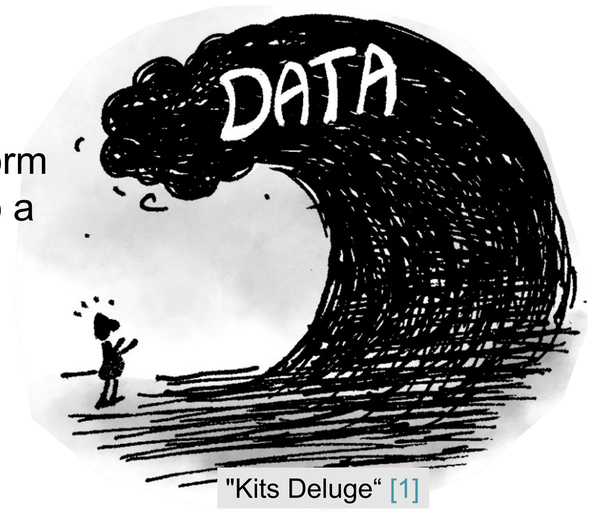
1 Introduction

At the Centre for Cellular Imaging, we develop and deliver smart microscopy solutions that integrate workflow automation with multiscale and multimodal imaging. These workflows generate diverse file formats that need to be combined into a holistic view of the sample

The primary challenge is managing the vast volumes of data (terabyte-scale) and the complex, multiscale, and multimodal nature of these imaging datasets. This involves transforming diverse (proprietary) file formats and image modalities into a standardized, cloud-compatible format to facilitate efficient data management, visualization, and analysis.

2 Objectives

Develop software tools to transform multimodal imaging datasets into a unified, cloud-compatible format, enhancing data management, visualization, and collaborative analysis in biological imaging.



"Kits Deluge" [1]

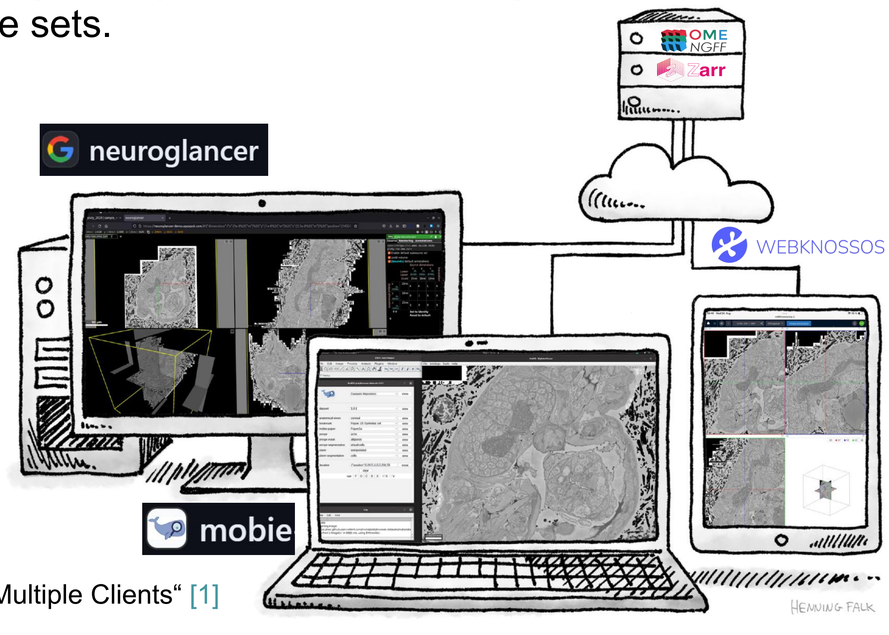
3 Methodology

Next-Gen File Formats: OME-Zarr for standardizing diverse imaging data, enabling cloud compatibility and computational efficiency.

Automated preprocessing: Implementing algorithms to enhance image quality and remove unnecessary data, optimizing storage efficiency.

Advanced visualizations tools: Utilizing OME-Zarr-compatible, community-driven tools (e.g., Napari, Webknossos, MoBiE/BDV, AICS 3D Cell Viewer, Neuroglancer) for enhanced microscopy data visualization. [2]

Collaborative Platforms: Ensuring constant interaction with users during the acquisition process (e.g., targeted imaging in Correlative Array Tomography workflows) and providing bioimage analysis services for the generated image sets.

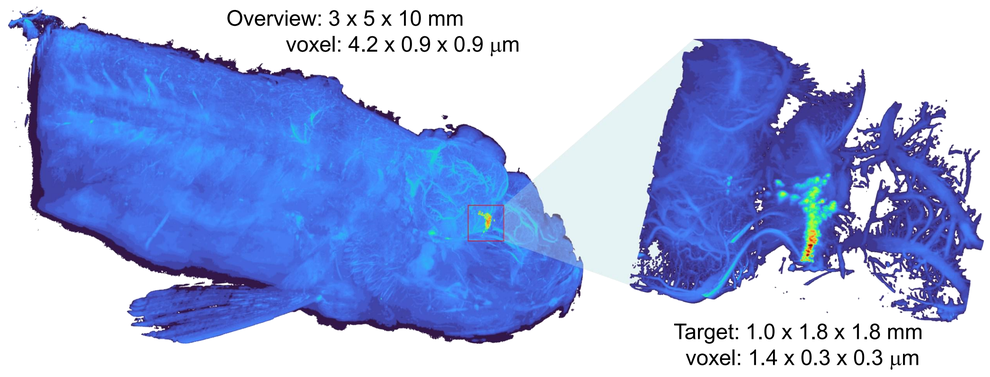


"Multiple Clients" [1]

4 Results & Case Studies

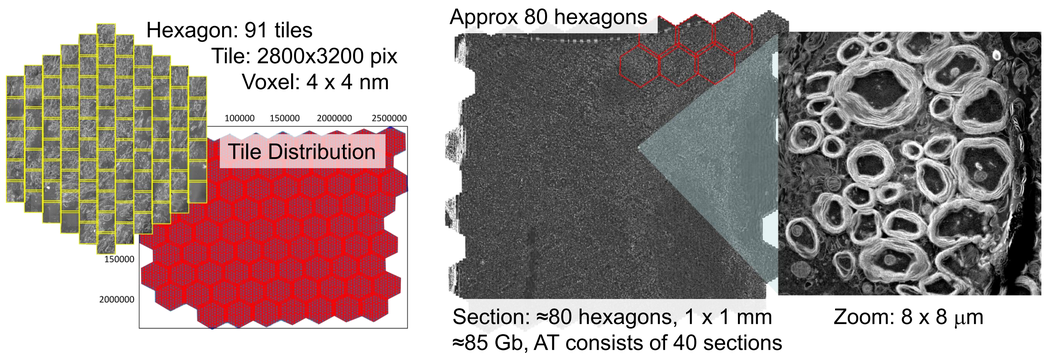
- Light-Sheet Microscopy:

Multiscale imaging of a zebrafish was performed using the Light Sheet 7 (ZEISS), capturing large full-sample volumes (left) and high-resolution targeted areas (right). The size of these datasets presents challenges in handling, preprocessing, visualization, and analysis. We preprocess and store the data in a correlated multiview format, enabling users to explore the dataset without relying on proprietary tools.



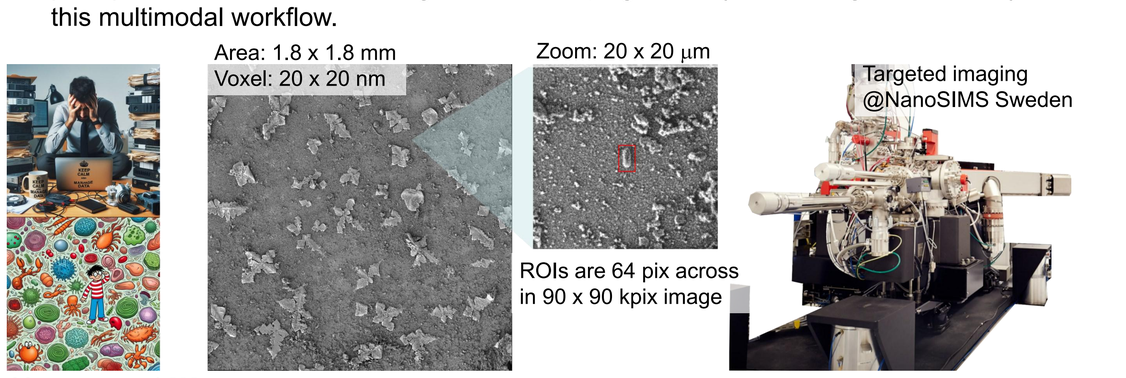
- Electron Microscopy – MultiSEM Array Tomography

The MultiSEM (ZEISS) generates data with 7200 image tiles per 1x1 mm section, organized into hexagonal arrangements that interconnect to form large, 85 GB sections at 4nm pixel resolution. To manage this complexity, we developed software tools to translate the intricate MultiSEM data structure into a single OME-Zarr format, enabling users to access and analyze the data without proprietary software. Our workflow also includes preprocessing steps for intensity corrections and basic axial alignment, with further improvements in alignment ongoing.



- NanoSIMS – Multimodal Targeted Imaging

We tackle the challenge of identifying rare or small objects within large sample areas—a true "needle in a haystack" scenario. Using SEM, we capture vast sample areas (mm²) at small voxel sizes (20-5 nm) to locate these targets. To enable collaboration, visualization, and annotation without large file transfers, we store data in the cloud-compatible OME-Zarr format. SEM images guide targeted NanoSIMS imaging, which, despite its high resolution, has a limited field of view. Our software tools translate SEM pixel positions to NanoSIMS stage coordinates, significantly enhancing the efficiency of this multimodal workflow.



AI generated [3]

https://www.gu.se/en/chemistry-molecular-biology/nanosims-sweden

5 Conclusion

We have developed tools to transform proprietary and large-scale image data into the cloud-compatible OME-Zarr format, facilitating efficient data management. Our collaboration and visualization frameworks leverage this data store to enhance remote and collaborative analysis. Examples are available in an open-access repository [4]. We also initiated collaborations with the community (e.g., ZMB-UZH [5, 6],) and industry (e.g., ZEISS .czi & Atlas 5, and Webknossos).

Future Work: Improve compatibility with ATLAS 5, ZEN Connect, and Thermo Scientific MAPS, and refine the UI for generating multimodal views from multiscale datasets.

6 References and Acknowledgements

[1] Images by Henning Falk, ©2022 NumFOCUS, is used under a CC BY 4.0 license;
[2] https://ngff.openmicroscopy.org/tools/; [3] Image generated using Microsoft Designer;
[4] https://github.com/CamachoDejay/ome-zarr-examples; [5] https://github.com/ZMB-UZH/ome-zarr-converters; [6] BioImage Archive S-BIAD1235;

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